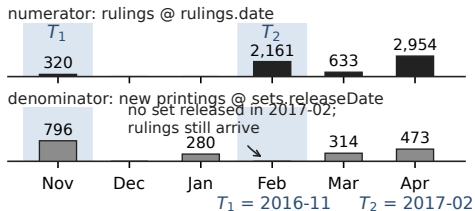


(a) one governed question, two as-of points

"ruling_intensity of card_games, as of T?"



correct under
as-of semantics

40.20%

REFUSE

320/796

missing-caliber

witness (2,161, 0):
no same-window denominator

evaluated LLM arms
(frozen runs)

0/8

X

6/8

X

returned 40.20% answered anyway

at T₂ the governed answer is not a number at all —
yet 6/8 evaluated LLM arms answered; the strongest
(governance-informed) arm computed 2,161/0 and returned "inf";
at the benign T₁, 0/8 arms returned the governed 40.20%

(b) the compile-time pipeline, traced on the same question

running example (panel a), traced:

① question

§2.1

$\langle q, T \rangle$: metric, scope, one window per metric
role, optionally a declared anchor $\bar{a}$

CARD-Q7: ruling_intensity, T=2017-02-15

② binding compilation

§2-3

pin $v = \text{ver}(T)$ (off-diagonal reads carry a mark);
derive α , caliber routing ρ ; guards OOV $\rightarrow$ AM $\rightarrow$ MC

$v = (\text{card_games}, v1)$
 $\alpha(\text{num}) = (\text{rulings.date}, \text{Feb})$
 $\alpha(\text{den}) = (\text{sets.releaseDate}, \text{Feb})$
probe $\mu_{\text{den}} = 0 \Rightarrow \text{MC(ii)}$ fires

③ disclosure gate

§4

maximal legal rewriting: grain frontier $U_{\min}$,
mask closure μ^* ; empty frontier $\Rightarrow$ DB

not reached: a binding guard fired first

④ decision + certificate

§5

ANSWER | REWRITE(q') | REFUSE(r), with
 $C = \langle q, T, \langle \alpha, v, \rho, \delta \rangle, \text{out} \rangle$; refusals carry a witness

REFUSE MC(ii), witness (2,161, 0)
sibling T₁=2016-11: ANSWER 40.20%

⑤ independent verifier

§5-6

$\text{Chk}(C, G_v, D, \text{ctx}) \rightarrow \text{ACCEPT} | \text{REJECT}$; every claim
re-derived from (G_v, D) ; zero modules shared with ①-④

ACCEPT (probe re-run on D: 0 rows)

versioned governed semantic layer G_v : anchors A_v , bindings β_v , caliber routing R_v ,
disclosure policies $\mathcal{D}$ + warehouse D (9 public DBs, 3.83M real rows)